



HopeDrop Blood Bank Management System

USER MANUAL

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Team: Group 50 - DevDynamos

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1. Introduction and System Overview

HopeDrop is a web-based Blood Bank Management System designed to modernise and digitalise the blood supply chain in Sri Lanka. It connects donors, blood camp organisers, hospital staff, doctors, and administrators on a single unified platform, eliminating the inefficiencies and delays caused by manual, paper-based processes.

This manual provides step-by-step instructions for users of every role.

1.1 System Roles

Role	Access Level	Primary Responsibilities
Donor	Restricted	Register, check eligibility, donate at camps, view history
Doctor	Restricted	Submit blood requests, track approvals, view hospital inventory
Blood Camp Organizer	Restricted	Create camps, manage registrations, dispatch blood to hospital
Inventory Staff (Inventor)	Restricted	Receive blood, quality check, submit inventory changes, manage expiry
Admin	Full	Manage all hospitals, doctors, inventory, requests, expiry alerts

1.2 Technology Platform

- Frontend: React 19 with Vite - accessible via any modern web browser
- Backend: Django 5.2 REST API
- Authentication: JWT (JSON Web Tokens) with 15-minute access token lifetime
- Database: SQLite (development) / PostgreSQL (production)

NOTE: HopeDrop works best on Google Chrome, Firefox, or Microsoft Edge. JavaScript must be enabled.

2. Getting Started - Accessing HopeDrop

Open your web browser and navigate to the HopeDrop application URL provided by your administrator.

The Landing Page contains:

- Navigation bar: Home, About Us, Services, Events, Blood Camps, Contact
- Hero section with a Donate Now call-to-action button
- Live blood stock information panel showing current availability by blood type
- Upcoming blood camp listings
- Login and Sign Up buttons in the top-right corner

2.1 Account Registration (New Users)

Navigate to /signup or click Sign Up from the navigation bar. The form collects:

Field	Details	Required?
Full Name	Your legal full name	Yes
Username	Max 10 characters, unique system identifier	Yes
Email	Valid email address (used for login)	Yes
NIC Number	12-digit or 9-digit+letter (Sri Lanka format)	Yes
Blood Group	A+, A-, B+, B-, AB+, AB-, O+, O-	Yes
Phone Number	Sri Lanka format (+94...)	Yes
Country	Select from dropdown	Yes
District	Select from dropdown based on country	Yes
Hospital	Your associated hospital	Optional
Role	donor (default), doctor, bloodcamp, Inventor	Yes
Password	Min 8 characters, not entirely numeric	Yes
Confirm Password	Must match Password	Yes

IMPORTANT: After registration, accounts default to the donor role. Doctor and Inventor roles may require admin activation.

2.2 Logging In

Navigate to /login. Enter your registered Email Address and Password, then click Login. You will be redirected to your role-specific dashboard:

Your Role	Redirected Dashboard
donor	/donor - Donor Dashboard
doctor	/doctor - Doctor Dashboard
bloodcamp	/bloodcamp - Blood Camp Organizer Dashboard
Inventor	/inventory - Inventory Staff Dashboard
admin	/admin - Admin Dashboard

2.3 Forgot Password

Click Forgot Password on the login page. Enter your registered email. You will receive a reset link by email. Click the link and set a new password on the Reset Password page (/reset-password).

WARNING: The reset link expires after a short period. If expired, request a new one.

3. Donor Portal

The Donor Portal gives blood donors complete control over their donation journey - from checking eligibility and registering for blood camps to viewing their full donation history and receiving personalised alerts.

3.1 Donor Dashboard (/donor)

After logging in, donors land on the Donor Dashboard which contains:

- Profile Summary - Name, blood group, total donations, and your unique QR code
- Next Eligible Donation Date - Countdown based on the 90-day Sri Lanka NBTS rule
- Donation History - Recent donation records with dates, hospitals, and statuses
- Upcoming Events - Blood camps you can register for
- Alerts and Notifications - System messages about eligibility, urgent requests, camps

3.2 Eligibility Checker (/donor/eligibility)

The eligibility checker tells you whether you are currently eligible to donate blood.

- If you have never donated: You are immediately eligible
- If you have donated before: The system calculates 90 days from your last donation date
- Green Eligible badge means you can donate now; red badge shows your next eligible date

NOTE: The 90-day minimum gap is based on Sri Lanka National Blood Transfusion Service (NBTS) standards.

3.3 QR Code

Every registered donor receives a unique QR code (UUID-based) linked to their profile. Your QR code can be found on your Donor Dashboard.

- Blood camp organisers scan it to register your arrival at a camp
- It identifies you uniquely without requiring physical documents
- Public profile accessible via /donor/scan/<your-qr-id>

3.4 Donation History and Certificates

The Donation History section lists every donation you have made:

Field	Description
Donation Date	Date of the donation
Hospital Name	Hospital where the donation was made
Blood Group	Blood type donated (snapshot at time of donation)
Units	Number of units donated
Status	Completed / Pending / Cancelled
Certificate ID	Unique UUID for your printable donation certificate

3.5 Blood Camp Registration

To register for an upcoming blood camp:

- Step 1: View Upcoming Events on your Donor Dashboard
- Step 2: Click Register on a camp of interest
- Step 3: Select your preferred appointment time if available

- Step 4: Confirm registration - you will appear in the camps donor list

Your registration status progresses through these stages:

Status	Meaning
registered	You have signed up for the camp
arrived	You have been checked in at the camp venue
screening	You are undergoing pre-donation medical screening
approved	Screening passed; you can proceed to donate
rejected	You did not pass screening (a reason is recorded)
donated	Your donation has been successfully collected

NOTE: You can only register once per camp. Duplicate registration is prevented by the system.

3.6 Notifications (/donor/notifications)

The Notifications page lists all system messages addressed to you. Types include:

- eligibility - Reminders when you become eligible to donate again
- urgent - Urgent blood requests matching your blood group
- camp - Announcements about upcoming blood camps
- info - General system information

Click on a notification to mark it as read. Unread notifications show a badge on the navigation bar.

4. Doctor Portal

Doctors use HopeDrop to submit blood requests for patients, monitor the approval process, and view real-time blood inventory at their hospital.

IMPORTANT: Doctor accounts are created and activated by the System Administrator. Contact your admin to set up credentials.

4.1 Doctor Dashboard (/doctor)

- Request Summary - Total submitted, pending, approved, and fulfilled requests
- Blood Inventory Panel - Current stock levels at your hospital by blood type
- Recent Requests - Table of most recent blood requests with status badges
- Analytics Charts - Visual breakdown of requests by status and blood type

4.2 Submitting a Blood Request

To request blood for a patient:

- Step 1: Click New Blood Request button on your dashboard
- Step 2: Fill in the request form (fields below)

Field	Description	Options
Blood Group	Required blood type	A+, A-, B+, B-, O+, O-, AB+, AB-
Units Requested	Number of blood units needed	Positive integer
Priority Level	Clinical urgency	NORMAL / HIGH / CRITICAL
Urgency	Operational urgency	NORMAL / HIGH / CRITICAL
Reason	Clinical justification	Free text

- Step 3: Click Submit Request
- Step 4: Your request immediately appears with status PENDING

4.3 Blood Request Lifecycle

Your request moves through the following approval workflow:

Status	Meaning	Action By
PENDING	Submitted, awaiting staff verification	Doctor
VERIFIED	Lab/staff confirmed the clinical need	Lab Staff
APPROVED	Admin approved blood allocation	Admin
IN_PROGRESS	Blood is being prepared for dispatch	Staff
FULFILLED	Blood has been delivered to the ward	Staff
REJECTED	Request denied (note recorded)	Admin/Staff
CANCELLED	Request cancelled by the doctor	Doctor

TIP: CRITICAL priority requests are highlighted prominently in the admin view to ensure faster processing.

5. Blood Camp Organizer Portal

Blood Camp Organisers manage the full lifecycle of donation camps - from creation and donor registration to blood collection and dispatch to hospitals.

5.1 Camp Dashboard (/bloodcamp)

- Your Camps - List of all camps you have created with status and statistics
- Upcoming Registrations - Donors who have registered for your next camp
- Collection Summary - Blood units collected per blood type
- Screening Queue - Donors currently in the screening process

5.2 Creating a Blood Camp

Click Create New Camp on your dashboard and fill in:

Field	Description
Title	Camp name/title (max 200 characters)
Date	Date of the camp
Start Time	Camp opening time
End Time	Camp closing time
Location	Physical venue address (max 255 characters)
Description	Optional additional information for donors
Destination Hospital	Hospital that will receive the collected blood

Camp status flow:

Status	Meaning
Upcoming	Camp is scheduled and open for donor registration
Ongoing	Camp is currently running on the day
Completed	Camp has finished; blood has been dispatched
Cancelled	Camp has been cancelled

5.3 Camp Day Workflow

- Step 1 - Donor Arrives: Scan the donors QR code and click Mark Arrived
- Step 2 - Screening: Send the donor to medical screening, click Send to Screening
- Step 3 - Approve or Reject: Click Approve (pass) or Reject (fail, record reason)
- Step 4 - Collection: Click Mark Donated to record the blood donation
 - A CampBloodCollection record is automatically created
 - Donors last_donation_date is updated
 - Donors total_donations counter increments

5.4 Dispatching Blood to Hospital

- Step 1: Go to Camp Collections to view all collected blood units
- Step 2: Click Dispatch All to mark all collections as in_transit
- Step 3: The destination hospital staff receive a notification to accept the blood

NOTE: Blood dispatched from a camp is tracked through the entire transit lifecycle until it reaches hospital inventory.

6. Inventory Staff / Lab Technician Portal

Inventory Staff (role: Inventor) manage the physical blood inventory at hospitals. They receive blood from camps, perform quality checks, and submit inventory change requests through a 4-stage approval workflow.

6.1 Staff Dashboard (/inventory)

- Pending Inventory Change Requests - Your submitted requests and their approval status
- Camp Blood Pending Receipt - Blood collections in transit awaiting your receipt
- Current Inventory Summary - Blood stock levels at your hospital
- Expiry Alerts - Units expiring soon requiring your action

6.2 Receiving Blood from a Camp

- Step 1: Navigate to Pending Camp Blood section
- Step 2: Find the incoming collection (status: in_transit)
- Step 3: Click Mark Received - status changes to received
- Step 4: Perform physical quality check in the lab
- Step 5: Click Verify / Quality Check
 - Pass: Status moves to quality_check then added_to_inventory
 - Reject: Status moves to rejected (record rejection reason)

6.3 Submitting Inventory Change Requests

All inventory changes must go through the 4-stage approval workflow. Navigate to Inventory then New Change Request:

Action Type	When to Use
ADD	Adding new blood units directly (e.g., direct donations)
ADJUST	Correcting quantity discrepancies
DELETE	Removing a blood unit record
MARK_DAMAGED	Recording a damaged blood unit
MARK_EXPIRED	Recording an expired blood unit

The 4-stage approval process:

- Stage 1 (PENDING_ADMIN_REVIEW): Request is submitted; admin reviews it
- Stage 2 (PENDING_QUALITY_CHECK): Admin approves; quality reviewer checks
- Stage 3 (PENDING_FINAL_ADMIN_APPROVAL): Quality passes; final admin sign-off
- Stage 4 (APPROVED_READY_TO_EXECUTE then EXECUTED): Request is executed

WARNING: A request can be REJECTED at any stage. You will receive a note explaining the reason.

6.4 Managing Expiry Alerts

Alert Type	Trigger Condition
expiring_soon	Blood unit expires within 7 days
expiring_critical	Blood unit expires within 3 days
expired	Blood unit has already passed its expiry date

To resolve an alert:

- Step 1: Navigate to Expiry Alerts section
- Step 2: Select the alert and choose a resolution action: used / discarded / transferred
- Step 3: Click Resolve - the alert is marked resolved with timestamp

7. Admin Portal

The Admin has full visibility and control over the entire HopeDrop system. Admin functions at /admin are protected and only accessible with the admin role.

7.1 Admin Dashboard (/admin)

- Total donors, doctors, active camps, blood requests
- National blood stock summary charts
- Recent blood requests requiring action
- Expiry alerts requiring resolution
- Pending inventory change requests

7.2 National Inventory Dashboard (/admin/inventory)

- View blood stock across all registered hospitals on one screen
- National live stock by blood type with colour-coded availability indicators
- Click on any hospital to drill into hospital-specific inventory detail
- Hospital Inventory Detail shows blood type, quantity, dates, status, and source

7.3 Hospital Management (/admin/hospitals)

- Add new hospitals with name, location, and contact information
- View all registered hospitals and their associated blood banks
- Deactivate hospitals as needed

7.4 Doctor and Staff Management (/admin/doctors)

- Create Doctor accounts: full name, email, specialisation, license number, hospital
- Activate or deactivate doctor accounts
- Manage Lab Staff / Inventory Officers: employee ID, designation, shift

7.5 Blood Request Management (/admin/requests)

- Filter by status: PENDING, VERIFIED, APPROVED, IN_PROGRESS, FULFILLED, REJECTED
- Review each request: blood group needed, units, urgency, reason
- Approve: allocate units and update status to APPROVED
- Reject: record a rejection note for the doctor

IMPORTANT: CRITICAL priority requests are shown at the top of the list and highlighted in red.

7.6 Camp Blood Tracking (/admin/camp-blood)

- View all CampBloodCollection records with current status
- Monitor transit progress from camp to hospital
- Identify collections stuck in transit

7.7 Expiry Alert Management (/admin/expiry-alerts)

- Filter by alert type: expiring_soon / expiring_critical / expired
- Resolve alerts by selecting: used / discarded / transferred
- View resolution history with timestamps

7.8 Donations Overview (/admin/donations)

- View every donation by date, donor, hospital, blood group, and units
- Full audit trail of all donations in the system

8. Public / Patient Features

The following features are accessible without logging in:

- Landing Page (/) - System overview, live blood stock, upcoming camps
- Blood Camps (/blood-camps) - List of upcoming and ongoing blood camps
- Events (/events) - Blood donation events and drives
- About Us (/about-us) - Information about HopeDrop and the team
- Services (/services) - Overview of services offered
- Contact (/contact) - Contact form to reach the admin team
- Public Donor QR Scan (/donor/scan/<qr-id>) - View a donors public profile
- Patient Dashboard (/patient) - Accessible to doctors and medical officers

9. Profile and Common Features

9.1 Updating Your Profile

After logging in, navigate to your dashboard profile section. You can update:

- Full Name and Phone Number
- District and Country
- Profile Photo (for donors and doctors)
- Associated Hospital

WARNING: NIC number, email, and username cannot be changed after registration. Contact your administrator for corrections.

9.2 Logging Out

Click the Logout button in the top-right corner or sidebar. This immediately invalidates your refresh token via token blacklisting, ensuring your session is securely terminated.

10. Troubleshooting and FAQ

Problem	Solution
Cannot log in	Check email/password. Use Forgot Password to reset. Ensure account is activated by admin.
Dashboard shows no data	Ensure the backend server is running. Check internet connection. Try refreshing.
Blood request stuck on PENDING	Requests are reviewed by lab staff and admin. CRITICAL requests are prioritised. Contact admin if pending >24h.
Cannot register for a camp	You may already be registered for that camp (duplicate registration is prevented).
QR code not scanning	Ensure camera has permission. Show QR code on a bright screen at arm length.
Expiry alert not resolving	Select a resolution action (used/discarded/transferred) before clicking Resolve.
Submitted wrong blood request	Contact admin immediately. It may be cancelled if not yet verified.
System shows ineligible to donate	HopeDrop enforces the 90-day NBTS standard. Check your last donation date.
Forgot to mark donor as donated	Navigate to the camp registration list, find the donor, update status to donated.
Password reset email not received	Check spam folder. Ensure you used the registered email address.

11. Glossary

Term	Definition
JWT	JSON Web Token - a secure, time-limited token used for authentication
NBTS	National Blood Transfusion Service (Sri Lanka)
QR Code	Quick Response code - a 2D barcode uniquely identifying each donor
Blood Inventory	The stock of blood units stored at a hospital blood bank
Expiry Alert	System notification when a blood unit is nearing or past its expiry date
Camp Registration	A donors formal sign-up for a specific blood donation camp
Change Request	A formal request to modify blood inventory, subject to 4-stage approval
Blood Group	Classification of blood: A+, A-, B+, B-, O+, O-, AB+, AB-
NIC	National Identity Card (Sri Lanka)
Donation Gap	Minimum 90-day interval required between donations (NBTS standard)
Inventory Officer	Staff member responsible for managing hospital blood inventory
Transit	The process of moving blood from a donation camp to a hospital
Role	The users function in the system: donor/doctor/admin/bloodcamp/Inventor